

1.

1.4 (2016 SP:11)

20.0 mL of 0.10 mol L⁻¹ hydrochloric acid is mixed with 20.0 mL of 0.10 mol L⁻¹ sodium hydroxide in a glass beaker. The volumes are measured using a 50.0 mL measuring cylinder. The temperature rise that occurred is measured and used to calculate the enthalpy change for the reaction. Which one of the following statements is correct?

- (a) Systematic error will be reduced by repeating the experiment several times and averaging the results.
- (b) Random error will be reduced by using a 20.0 mL graduated pipette instead of the 50.0 mL measuring cylinder.
- (c) Random error will be reduced by insulating the beaker.
- (d) Systematic error will be increased by doubling the volume of solution.

2.

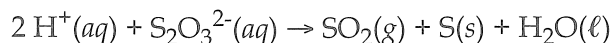
1.4 (2016:06)

Which one of the following sets of titres indicates a systematic error if the actual volume being measured is 85.2 mL?

- (a) 85.1 mL, 85.1 mL, 85.3 mL, 85.5 mL
- (b) 65.2 mL, 75.2 mL, 85.2 mL, 95.2 mL
- (c) 85.2 mL, 85.3 mL, 85.1 mL, 85.1 mL
- (d) 87.3 mL, 86.9 mL, 89.1 mL, 88.2 mL

Questions (2017:22) to (2017:25) refer to the following information.

Some chemistry students were investigating the relationship between concentration and rate of reaction. In the investigation, different concentrations of hydrochloric acid were added to a sodium thiosulfate solution to produce solid sulfur. This reaction was represented by the following equation.



A piece of paper with a cross drawn on it was placed under the reaction vessel. The time taken for the cross to disappear due to the formation of the precipitate was measured.

3. 1.3 (2017:22)

The independent variable was the

- (a) time taken for the cross to disappear.
- (b) total volume of the mixture.
- (c) rate of reaction.
- (d) concentration of hydrochloric acid.

4. 1.3 (2017:23)

The type of data collected and the source of data are best characterised as

- (a) qualitative and primary.
- (b) qualitative and secondary.
- (c) quantitative and primary.
- (d) quantitative and secondary.

5. 1.3 (2017:24)

When a number of laboratory groups pooled their data, one group's results were consistently higher than those of the others. This is an example of

- (a) a systematic error.
- (b) not enough trials.
- (c) a random error.
- (d) uncertainty.

6.

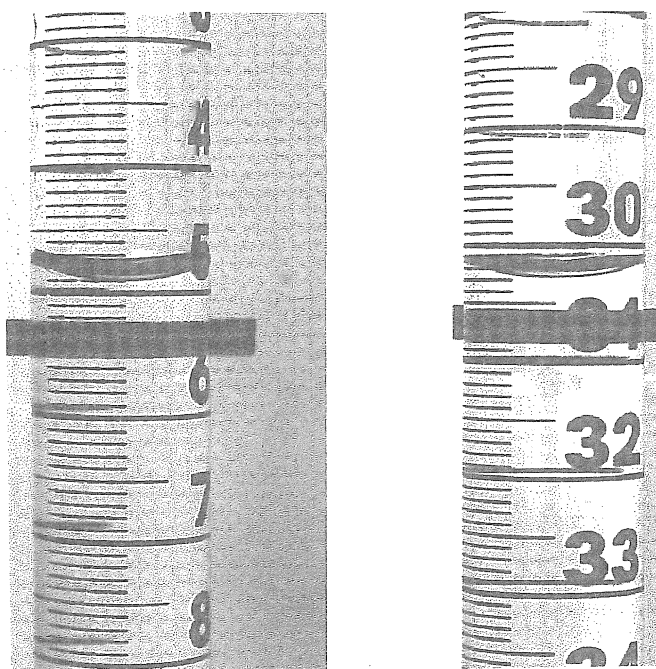
1.3 (2017:25)

One group chose to have its members take turns observing and timing the cross disappearing. This was poor methodology because

- (a) it could make the data invalid.
- (b) it introduced a possible systematic error.
- (c) more trials would be needed to produce better results.
- (d) the data would be less reliable.

7.

1.3 (2018:12)



The photographs above show a Class A burette before (left) and after (right) a titration. Use these photographs to determine the titre volume used in this titration.

- (a) 25.3 ± 0.05 mL
- (b) 25.33 ± 0.05 mL
- (c) 25.39 ± 0.10 mL
- (d) 26.6 ± 0.1 mL

8.

1.3 (2018:13)

The table below shows the volumes added from a burette during a titration.

Titre (mL)					
1	2	3	4	5	6
19.23	19.94	19.98	19.94	20.02	19.94

What value should be used in the titration calculations?

- (a) 19.84 mL
- (b) 19.94 mL
- (c) 19.95 mL
- (d) 19.96 mL

9.

1.2 (2018:16)

Chemists must act ethically when conducting research. Which of the following statements relate to **ethical** behaviour?

- (i) Chemists calibrate all of their instruments with primary standards.
 - (ii) Chemists give due credit to all contributors to an investigation in their written reports.
 - (iii) Chemists record their experimental results accurately and without alteration to fit their prediction.
 - (iv) Chemists dispose of their waste materials carefully, especially those containing heavy metals.
 - (v) Chemists ensure that there is only one independent variable in every experiment they perform.
 - (vi) Chemists declare any conflict of interest relevant to their investigation.
- (a) i, ii and iii only
 - (b) ii, iv and v only
 - (c) ii, iii, iv and vi only
 - (d) i, iii, iv, v and vi only

The following information relates to Questions (2019:19) and (2019:20).

A group of Year 12 Chemistry students wanted to know whether increasing ocean acidity increases the rate at which sea shells, CaCO_3 , dissolve. They went to a beach to collect seawater and sea shells. In their school laboratory they crushed the sea shells and added 2.00 g of the resulting powder to five clean 250 mL beakers, each of which had been placed on top of its own electronic balance.

They split the seawater into five portions and bubbled carbon dioxide gas into four of the portions for different amounts of time. This gave the students 'natural' seawater plus four seawater samples of different pH. The various seawaters (150 mL portions) were then added to the beakers, with the weight of each beaker and its contents being recorded at timed intervals.

10. 1.1 (2019:19)

Which one of the following proposes a suitable hypothesis for the investigation?

- (a) As the seawater becomes more acidic, the sea shell powder will dissolve faster.
- (b) The sea shell powder will dissolve fastest in the most acidic seawater.
- (c) Adding carbon dioxide to seawater changes the pH of the seawater.
- (d) More of the sea shell powder will dissolve as time progresses.

11. 1.3 (2019:20)

Which one of the following pairs of statements on the validity and reliability of the investigation is correct?

	Validity	Reliability
(a)	It is valid because the investigation allows them to determine if seawater pH affects the rate of sea shell dissolution.	It is reliable because the trials were performed in a laboratory.
(b)	It is not valid because the investigation was simulated in a laboratory and not performed in a real ocean.	It is not reliable because only one trial was performed at each different pH value.
(c)	It is not valid because the investigation was simulated in a laboratory and not performed in a real ocean.	It is reliable because trials were performed at five different pH values.
(d)	It is valid because the investigation allows them to determine if seawater pH affects the rate of sea shell dissolution.	Its reliability could be improved by conducting multiple trials at each different pH value.

12.

1.3 (2020:25)

A chemist performed an acid-base titration. The acid was in a burette and a pipette was used to deliver a known quantity of the base into a conical flask. Which of the following gives the final rinse solution for each of these pieces of equipment?

	Final rinse solution		
	Burette	Pipette	Conical flask
	(a) acid	water	base
	(b) acid	base	water
	(c) water	base	water
	(d) water	water	base

13.

1.2 (2021:17)

A chemist performed a series of titrations and published the results in a scientific journal. From the point of view of the chemist, the titration data is

- (a) primary.
- (b) secondary.
- (c) personal.
- (d) investigative.

14.

1.4 (2021:18)

An example of a random error in a titration is

- (a) reading solution volumes to the bottom of the meniscus.
- (b) a gas bubble in the burette tap that comes out during a titration.
- (c) calculating the concentration of the primary standard incorrectly.
- (d) rinsing down the sides of the conical flasks during titrations.

- 18.(2020:09) This is the standard structure of an alkyl (the straight chain) benzyl (the benzene ring) sulfonate (the negatively charged SO_3^-) – it is a detergent. The answer is (a).
- 19.(2020:22) Soaps form precipitates with calcium and magnesium ions dissolved in water. To avoid this problem, we use detergents because they do not form precipitates. The answer is (b).
- 20.(2020:23) In soap, for example, sodium stearate, the stearate ion is negatively charged. The long carbon chain is attracted to the grease by dispersion forces to form a micelle leaving the ion $-\text{COO}^-$ exposed to interact with water molecules. The answer is (c).
- 21.(2021:04) Use of polymers is related to chemical properties. Length determines dispersion forces and melting-point, functional groups can determine strength and crosslinking occurs with S-S and hydrogen bonds between H and O but not between H and H. The answer is a.
- 22.(2021:05) The strength of nylon is due to the interaction between NH and CO – it is a hydrogen bond. The answer is c.
- 23.(2021:06) There are 12 principles of green chemistry but you do not need to know them by rote. You should understand the idea of green chemistry with reference to ethanol and biodiesel. *Scientific knowledge can be used to design alternative chemical synthesis pathways, taking into account sustainability, local resources, economics and environmental impacts (green chemistry), including the production of ethanol and biodiesel.* The answer is a.
- 24.(2021:07) Alpha amino acids exist, not as shown in the Data Book, but as zwitterions. In acidic conditions the carboxylate group will become protonated. The answer is b.
- 25.(2021:11) Alpha amino acids have an amino group, an alpha carbon and a COOH group along with a side chain. The answer is a.

Chapter 6: Science Enquiry Skills

- 1.(2016 SP:11-b) Systematic error is caused by faulty equipment or faulty methodology and cannot be reduced by repeating an experiment. A pipette offers more sig figs and will lower random error. Answer is 'b'.
- 2.(2016:06-d) A systematic error can be identified by data that is precise (many values close together) but wrong. The answer is 'd'.
- 3.(2017:22-d) The concentration of HCl is varied and the effect on the other variables are measured. This makes it the independent variable. The answer is 'd'.
- 4.(2017:23-c) The data involved numbers (quantitative) and was collected by the experimenter (thus it was primary data). The answer is 'c'.
- 5.(2017:24-a) The data was consistently wrong which means it was not random error. It was systematic caused by a wrong technique or poor equipment. The answer is 'a'.
- 6.(2017:25-d) The technique would mean there were differing methods used and the data would be less reliable. The answer is 'd'.
- 7.(2018:12-c) A burette can be read to two decimal places and, unlike most glassware, zero is at the top. The error is half a graduation (0.05 mL). Errors are added when data is added or subtracted. So the error is 0.10 mL. The answer is 'c'.
- 8.(2018:13-d) Result 1 appears to be an outlier. It differs from the rest by 0.7 mL – a large amount. Thus, the remaining five values are averaged. The answer is 19.96 mL. The answer is 'd'.
- 9.(2018:16-c) Ethics have not previously been in a SCSA ATAR Chemistry examination. Items (i) and (v) are related to experimental design. The remainder concern doing research fairly and honestly. These relate to ethics. The answer is 'c'.
- 10.(2019:19-a) A hypothesis must relate to empirical data relating a dependent variable and an independent variable and be testable. The only answer to meet these criteria is 'a'.
- 11.(2019:20-d) This question involves understanding of the syllabus terms 'reliability' and 'validity'. Does the test measure what it is supposed to measure and are the results consistent? The answer is 'd'.
- 12.(2020:25) For the burette and pipette the 'solution to be used' is the final rinse and the conical flask is rinsed with distilled water. The answer is (b).
- 13.(2021:17) The scientist personally obtained the data so it is primary data. The answer is a.
- 14.(2021:18) Answer 'a' is good technique unless the liquid is like permanganate solution where we read from the top. Answer 'c' is a mistake – not a random error. Answer 'd' is good technique. The answer is b.

Chapter 7: Chemical Equilibrium

- 1.(2016 SP:34) a) $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq})$
 b) $\text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$
 c) $\text{H}_3\text{O}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\ell)$
 By lowering the pH more $\text{H}^+(\text{aq})$ is available to convert carbonate ion to hydrogencarbonate, lowering the $[\text{CO}_3^{2-}]$ and so limiting the ability of coral to form calcium carbonate reef.
- 2.(2016:41) a) $K = \frac{p\text{NO}_2(\text{g})^2}{p\text{N}_2(\text{g})p\text{O}_2(\text{g})^2}$ (preferred) or $K = \frac{[\text{NO}_2]^2}{[\text{N}_2][\text{O}_2]^2}$
 b) No change in K